

SET 8: SURVEY RESPONSES ANALYSIS

Dataset: Survey Responses

Department	AIDS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
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Dataset: Survey Responses

Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

PROGRAM 1: Grouped Bar Chart of Question 1 Answers

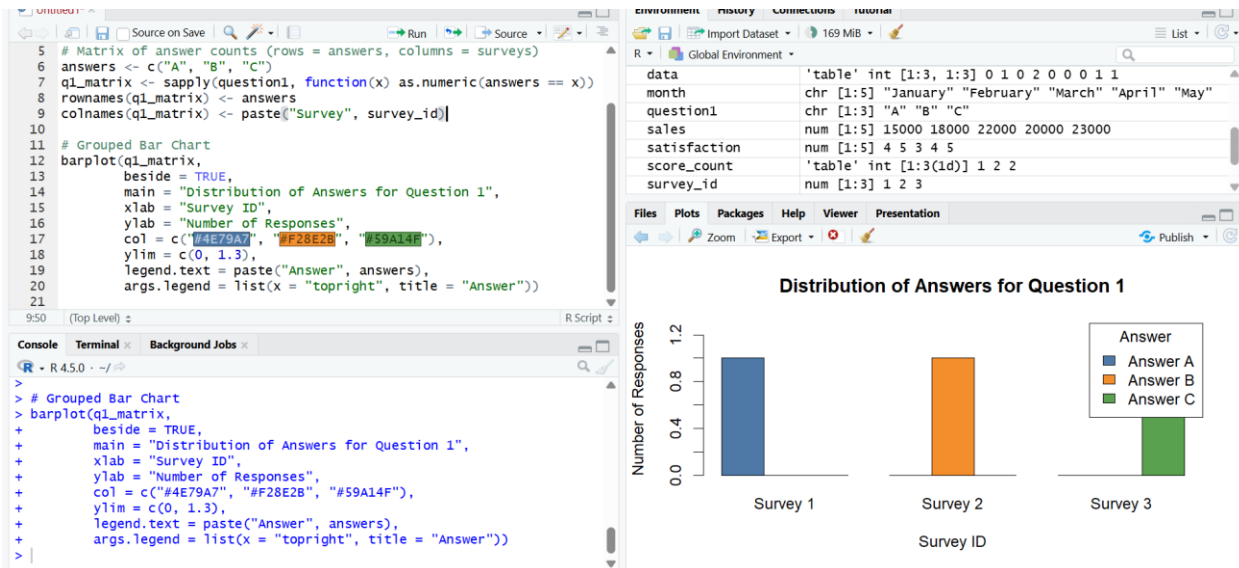
Question: Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.

```
# Dataset
survey_id <- c(1, 2, 3)
question1 <- c("A", "B", "C")

# Matrix of answer counts (rows = answers, columns = surveys)
answers <- c("A", "B", "C")
q1_matrix <- sapply(question1, function(x) as.numeric(answers == x))
rownames(q1_matrix) <- answers
colnames(q1_matrix) <- paste("Survey", survey_id)

# Grouped Bar Chart
barplot(q1_matrix,
        beside = TRUE,
        main = "Distribution of Answers for Question 1",
        xlab = "Survey ID",
        ylab = "Number of Responses",
        col = c("#4E79A7", "#F28E2B", "#59A14F"),
        ylim = c(0, 1.3),
        legend.text = paste("Answer", answers),
        args.legend = list(x = "topright", title = "Answer"))
```

Output 1: Grouped Bar Chart for Question 1



Observation: Each survey selected a different answer for Question 1 — Survey 1 chose A, Survey 2 chose B, and Survey 3 chose C. Every option therefore received exactly one response, and no respondent chose D.

PROGRAM 2: Stacked Bar Chart of All Questions

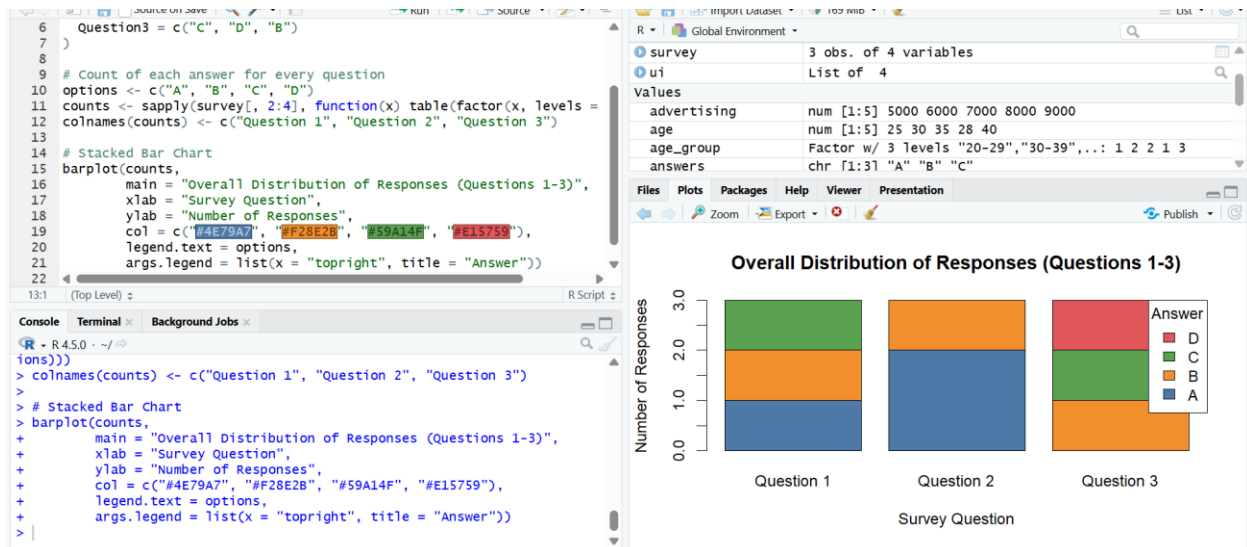
Question: Generate a stacked bar chart to represent the overall distribution of responses for all three questions.

```
# Dataset
survey <- data.frame(
  Survey_ID = c(1, 2, 3),
  Question1 = c("A", "B", "C"),
  Question2 = c("B", "A", "A"),
  Question3 = c("C", "D", "B")
)

# Count of each answer for every question
options <- c("A", "B", "C", "D")
counts <- sapply(survey[, 2:4], function(x) table(factor(x, levels =
options)))
colnames(counts) <- c("Question 1", "Question 2", "Question 3")

# Stacked Bar Chart
barplot(counts,
  main = "Overall Distribution of Responses (Questions 1-3)",
  xlab = "Survey Question",
  ylab = "Number of Responses",
  col = c("#4E79A7", "#F28E2B", "#59A14F", "#E15759"),
  legend.text = options,
  args.legend = list(x = "topright", title = "Answer"))
```

Output 2: Stacked Bar Chart for Questions 1–3



Observation: Each bar totals three responses. Question 1 has one each of A, B and C; Question 2 is dominated by answer A (2 responses) with one B; Question 3 has one each of B, C and D.

PROGRAM 3: Survey Response Table

Question: Build a table to show the survey response data for each survey. Label the table elements.

C

Output 3: Survey Response Table

The R code in the console creates a data frame for survey responses. The resulting table is displayed in the console output, showing the Survey ID, Question 1, Question 2, and Question 3 for three different surveys.

Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

Observation: The table lists the answers given by each of the three surveys to Questions 1, 2 and 3, with Survey ID as the row identifier.

PROGRAM 4: Interactive Tableau Dashboard

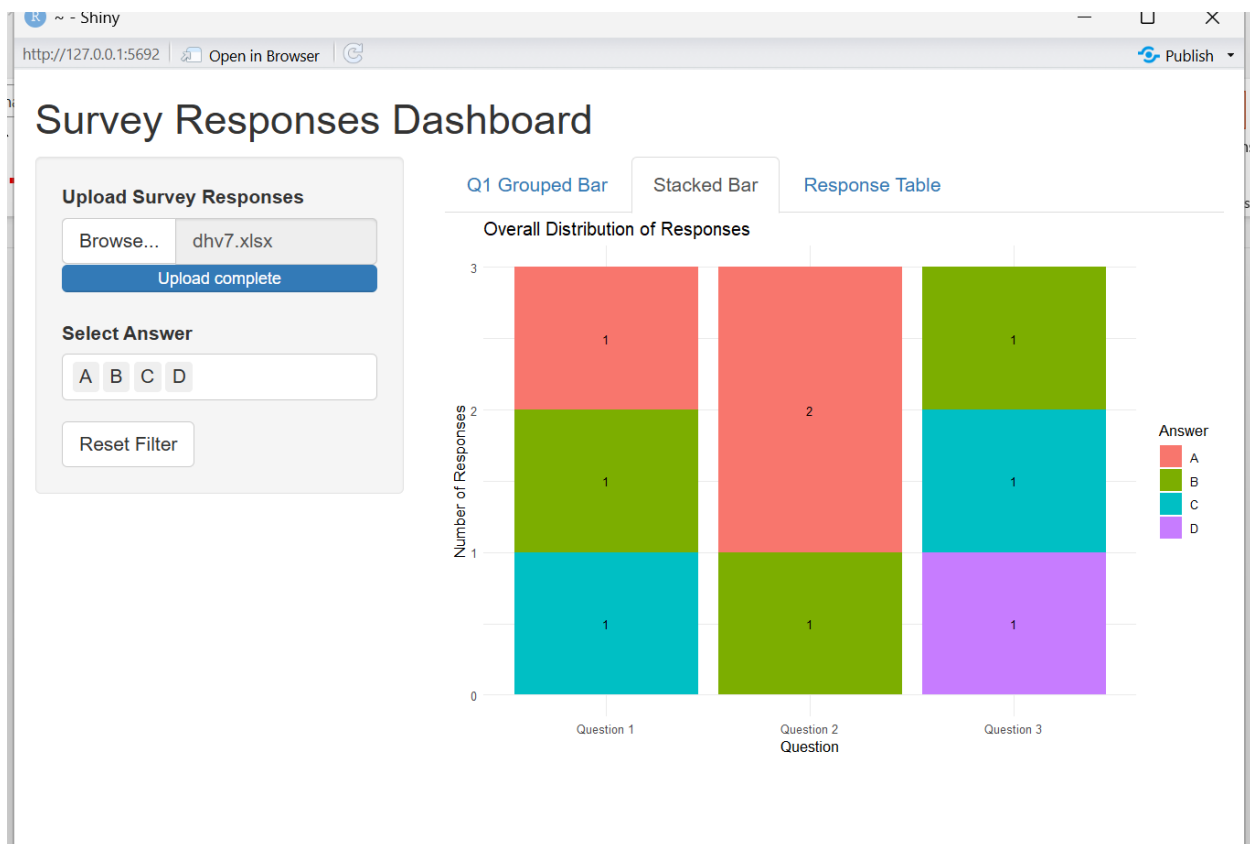
Question: Develop a Tableau dashboard combining the grouped bar chart, stacked bar chart, and the table for interactive exploration of survey responses.

Tableau Steps

1. Open Tableau and connect to the Survey Responses dataset (Excel/CSV).

2. Select Data > Pivot (or use Pivot in the Data Source tab) to convert Question 1, Question 2 and Question 3 into two fields: Question and Answer.
3. Create a new worksheet named “Q1 Grouped Bar”. Filter Question to Question 1.
4. Drag Survey ID to Columns and Answer to Columns (next to it) so bars are grouped; drag Number of Records (CNT) to Rows; place Answer on Color and select Bar in the Marks card.
5. Add the title “Distribution of Answers for Question 1”, label the axes (Survey ID, Number of Responses) and show mark labels.
6. Create a second worksheet named “Stacked Bar”. Drag Question to Columns, Number of Records to Rows and Answer to Color; Tableau stacks the bars automatically.
7. Add the title “Overall Distribution of Responses”, show the Answer legend and enable mark labels.
8. Create a third worksheet named “Response Table”. Drag Survey ID to Rows, Question to Columns and Answer to Text (or use the original un-pivoted data with Question 1–3 as columns).
9. Create a new Dashboard, set the size to Automatic, and drag in the three worksheets.
10. Add the title “Survey Responses Dashboard”.
11. Select the stacked bar chart, click the “Use as Filter” icon, and add an Answer quick filter (Show Filter) so selections update all views.
12. Test the dashboard by clicking an answer segment or choosing an Answer value, and check that the grouped bar chart and table update.

Output 4: Survey Responses Dashboard



Observation: The Tableau dashboard combines the grouped bar chart for Question 1, the stacked bar chart

for all questions and the survey response table, allowing interactive exploration of the survey responses through filters and click-to-highlight actions.

Procedure

1. Create the survey response dataset in R.
2. Compute the Question 1 answer counts per survey and draw the grouped bar chart using `barplot()` with `beside = TRUE`.
3. Count the answers (A–D) for each question using `table()` and draw the stacked bar chart using `barplot()`.
4. Create a data frame containing Survey ID and Questions 1–3 with labelled columns and display it as a table.
5. Open Tableau, import the Survey Responses dataset and pivot the question columns.
6. Create the grouped bar chart, stacked bar chart and response table as separate worksheets.
7. Combine the three worksheets into a dashboard, add a title, and enable filters.
8. Test the dashboard interactivity.

Result

The Survey Responses data was successfully visualized using a grouped bar chart, a stacked bar chart, a labelled response table, and an interactive Tableau dashboard.